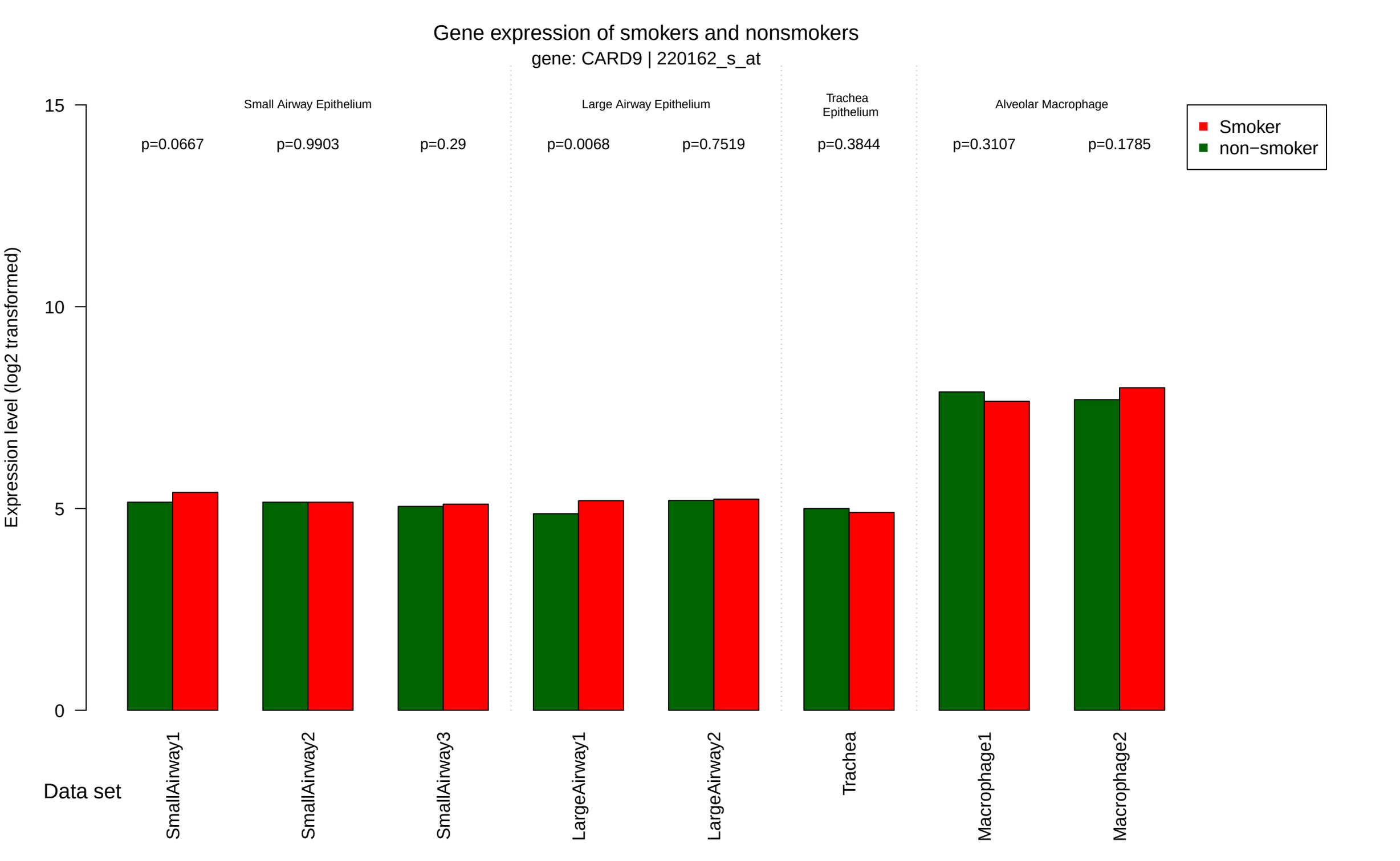
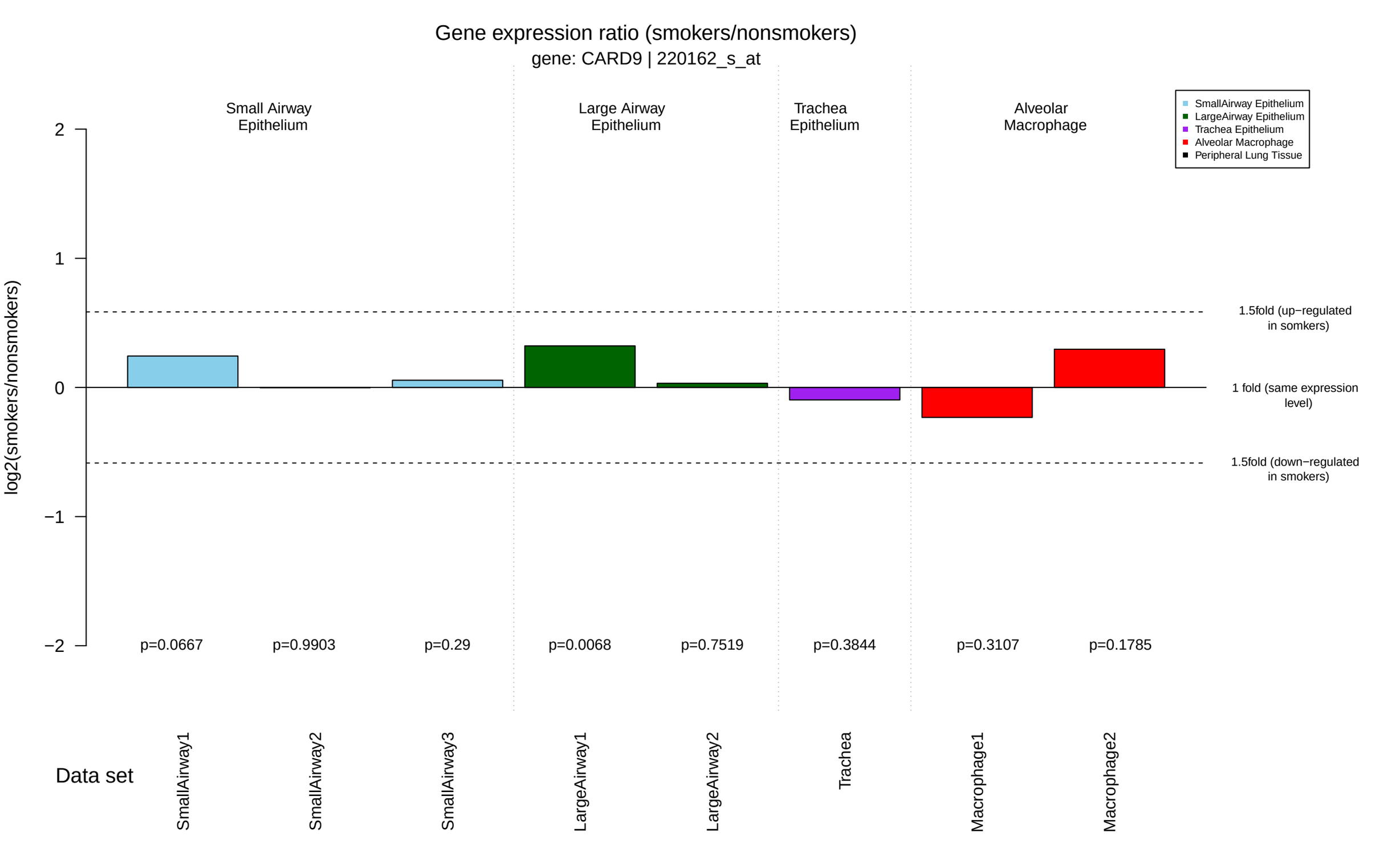
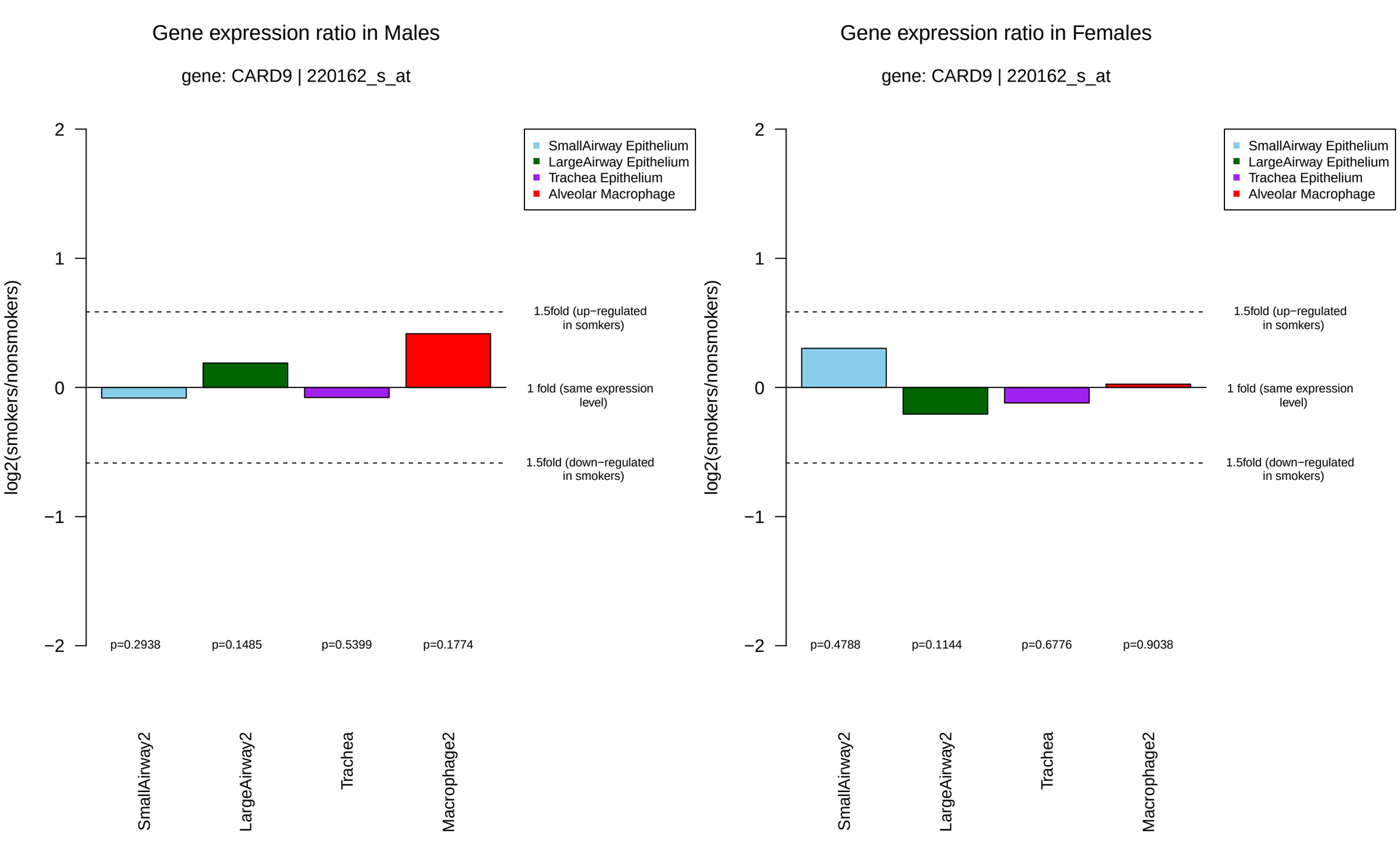
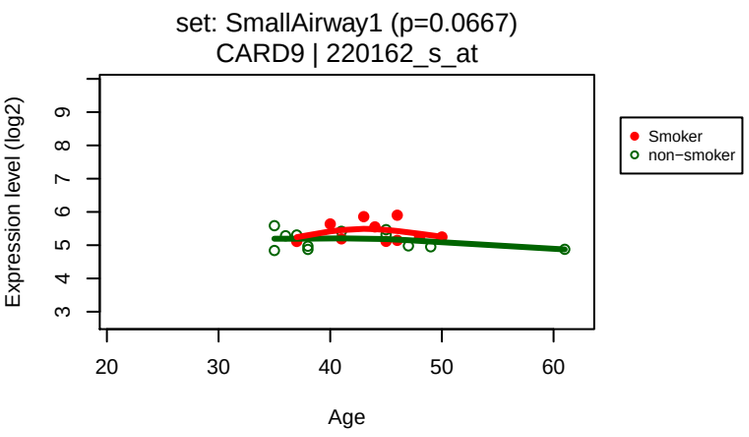
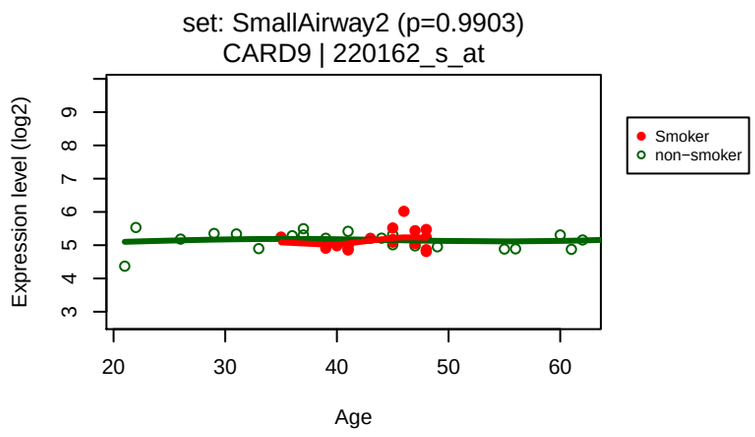
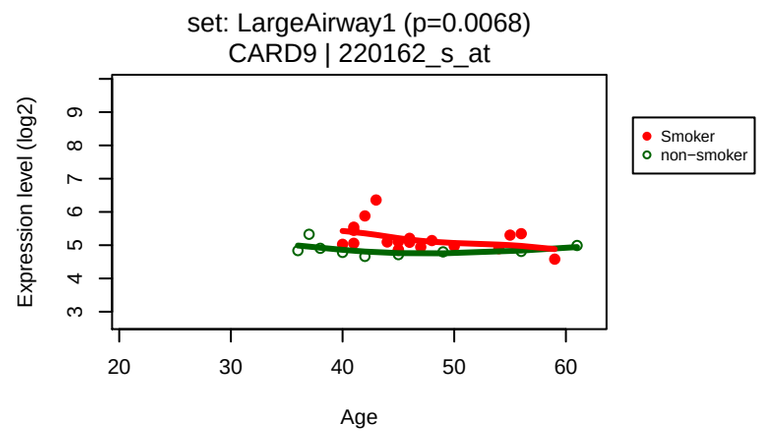

Gene expression of smokers and nonsmokers
gene: CARD9 | 220162_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0667
p=0.9903
p=0.29
p=0.0068
p=0.7519
p=0.3844
p=0.3107
p=0.1785
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

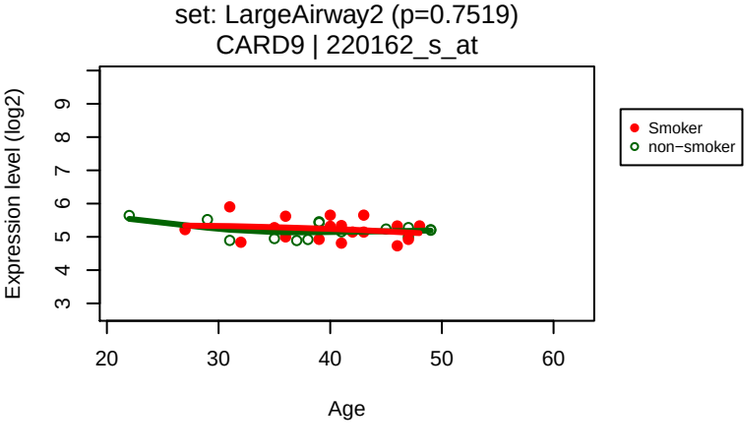

Gene expression ratio (smokers/nonsmokers)
gene: CARD9 | 220162_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0667
p=0.9903
p=0.29
p=0.0068
p=0.7519
p=0.3844
p=0.3107
p=0.1785
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

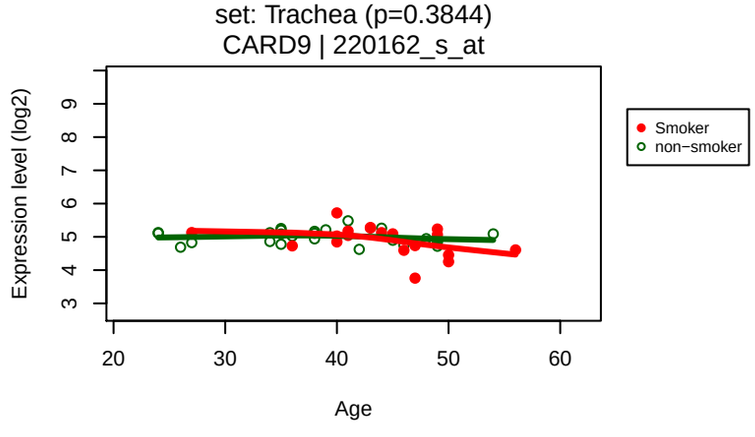

Gene expression ratio in Males
gene: CARD9 | 220162_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2938
p=0.1485
p=0.5399
p=0.1774
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CARD9 | 220162_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4788
p=0.1144
p=0.6776
p=0.9038
SmallAirway2
LargeAirway2
Trachea
Macrophage2

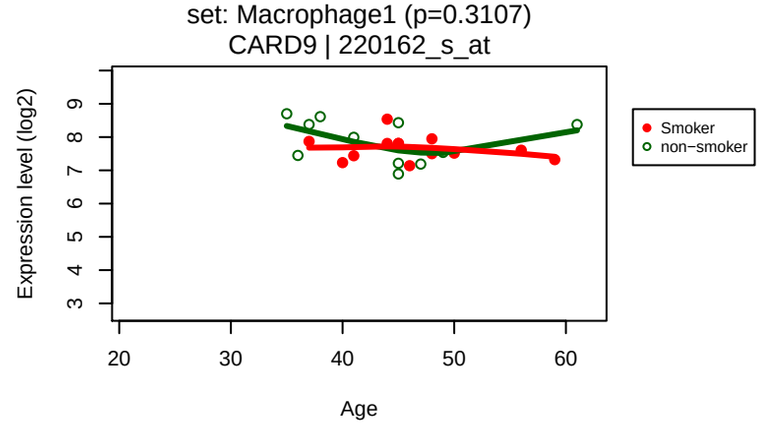

set: SmallAirway1 (p=0.0667)
CARD9 | 220162_s_at
Expression level (log2)
Age
Smoker
non−smoker

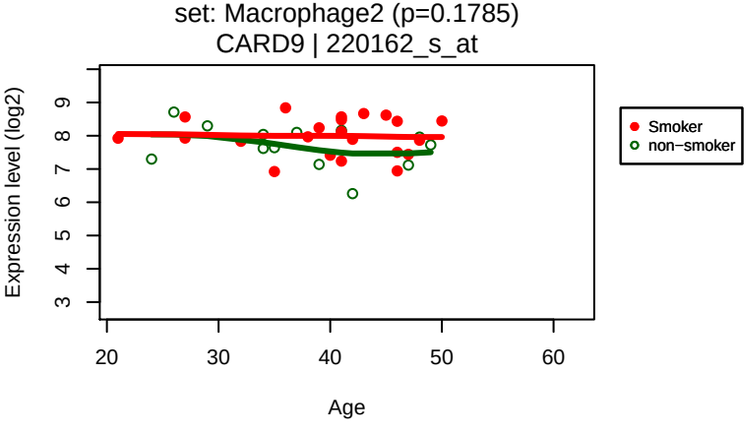

set: SmallAirway2 (p=0.9903)
CARD9 | 220162_s_at
Expression level (log2)
Age
Smoker
non−smoker

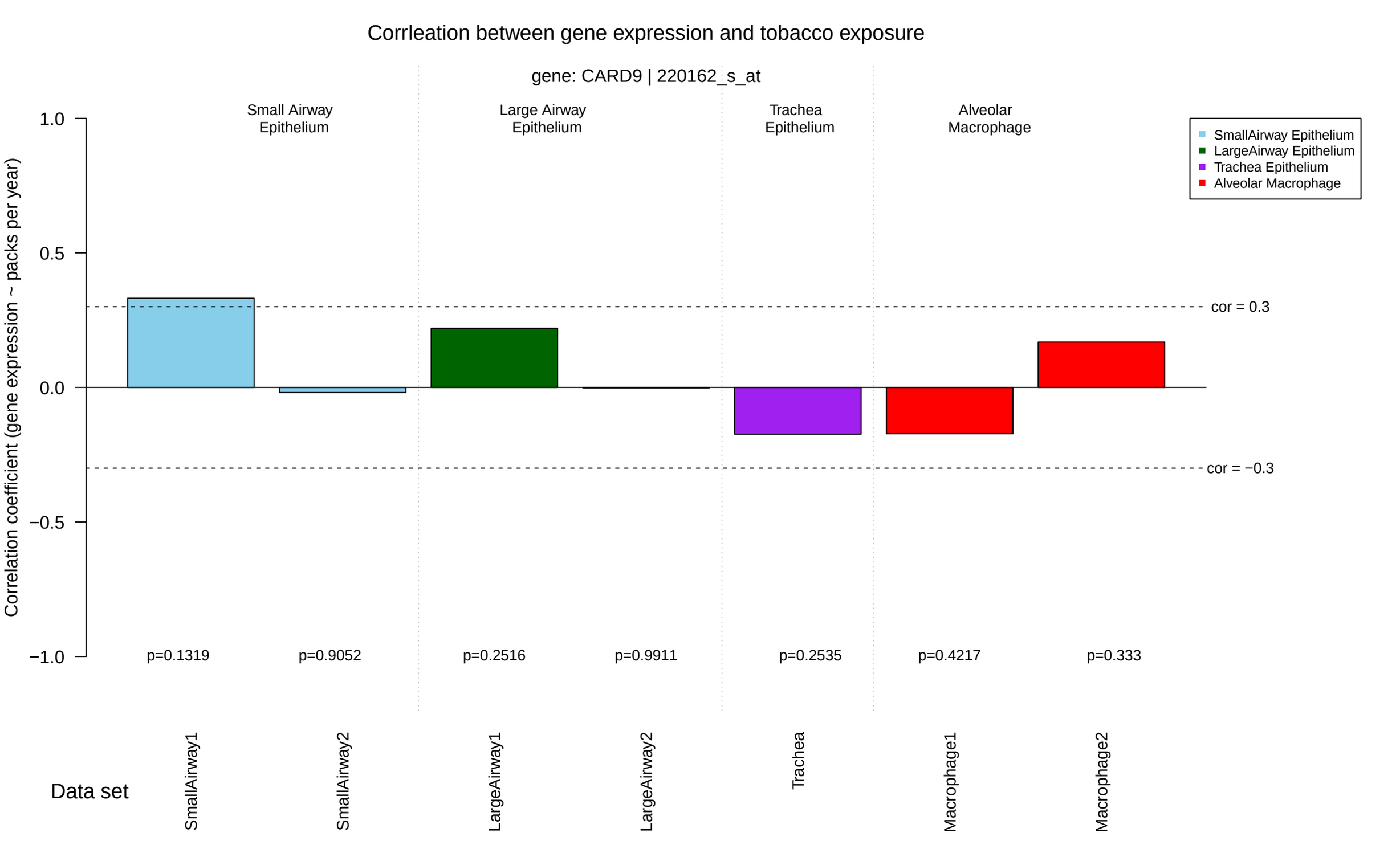

set: LargeAirway1 (p=0.0068)
CARD9 | 220162_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7519)
CARD9 | 220162_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3844)
CARD9 | 220162_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3107)
CARD9 | 220162_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1785)
CARD9 | 220162_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CARD9 | 220162_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1319
p=0.9052
p=0.2516
p=0.9911
p=0.2535
p=0.4217
p=0.333
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2